

Development Guide

Prerequisites

- JDK 21
- Maven 3.9+
- Node.js + Angular CLI (frontend)
- PostgreSQL with DB `bankone` and user `bankone_user`
- Optional: Open Liberty under `~/tools/wlp` (server name `bankone`)

Backend

```
cd "BankOne/BankOne-BackEnd"
# Embedded
mvn spring-boot:run
# Liberty (recommended) – rebuilds WAR, always enables JDWP :7777
./scripts/redeploy-liberty.sh
# Attach IntelliJ: Remote JVM Debug → localhost:7777
# Disable debug: LIBERTY_DEBUG=0 ./scripts/redeploy-liberty.sh
```

Local Kafka + Mailpit (notifications)

```
cd "BankOne"
docker compose up -d kafka mailpit
# Mailpit UI: http://localhost:8025
# Kafka: localhost:9092
```

Load-test data (optional)

```
psql -h localhost -U bankone_user -d bankone \
-f scripts/seed-loadtest-10k.sql
# Search UI for LoadTest; filter accounts by branch 9999
```

Key config: `BankOne-BackEnd/src/main/resources/application.properties`

Property	Purpose
<code>spring.datasource.*</code>	PostgreSQL connection
<code>spring.jpa.hibernate.ddl-auto=update</code>	Schema sync
<code>spring.sql.init.mode=always</code>	Runs <code>schema.sql</code>
<code>server.port=8080</code>	Embedded only
<code>jwt.secret</code> / <code>jwt.expiration</code>	JWT signing
<code>app.kafka.*</code> / <code>KAFKA_*</code>	Kafka notifications
<code>app.mail.*</code> / <code>MAIL_*</code>	SMTP or SendGrid transport

After Liberty redeploy, smoke login expects `admin / Admin@123`.

Frontend

```
cd "BankOne/BankOne-Frontend"
npm install # first time
ng serve
```

Set API host in `src/app/core/config/api-config.ts` (currently points at Liberty / LAN host on port 9080).

Coding conventions

Backend

- Package by domain: `com.bankone.{module}.{controller|service|repository|entity|dto|...}`
- Prefer service interface + `*Impl`
- Search via JPA `Specification` classes
- Map API responses through DTOs for accounts/policies; customers currently return entity (known smell)
- Security roles: `ROLE_` prefix handled by Spring; matchers use `ADMIN`, `EMPLOYEE`, `MANAGER`

Frontend

- Feature folders under `src/app/features/`
- Shared HTTP in `src/app/core/services/`
- Models in `src/app/core/models/`
- Bearer token via `auth.interceptor.ts`
- Dialogs as standalone components (e.g. `opening-deposit-dialog/`)
- Shared list footer: `app-list-pagination` (First/Prev/Next/Last + Go to page; 1-based UI → 0-based API page)

Teach-while-building

Project rule (`.cursor/rules/teach-while-building.mdc`): unless the user says to apply code, explain snippets for IntelliJ paste. Documentation sync still applies when code lands.

Documentation sync checklist (every change)

1. Update module doc under `docs/MODULES/`
2. Update API / schema / sequences / call flow if surfaces changed
3. Append `docs/CHANGELOG.md`
4. Refresh Mermaid diagrams when structure changes

Common pitfalls

Symptom	Likely cause
405 on new GET endpoint	Stale Liberty WAR — redeploy
Opening CURRENT fails on deposit	Policy requires min opening (seed CURRENT/INR = 5000)
CORS errors from phone/LAN	Firewall / AP isolation; CORS already allows <code>192.168.0.4</code>
Roles missing after login	Roles not in JWT — check <code>user_roles</code> and filter reload
Email shows <code>Customer@hash</code>	Do not concatenate <code>Customer</code> entity — use <code>customerLabel</code>
Render mail timeout on SMTP	Use <code>MAIL_TRANSPORT=sendgrid</code> (HTTPS); outbound SMTP blocked
Render Docker CACHED old WAR	Manual Deploy → Clear build cache & deploy

Suggested next build slices

Product / hardening (separate lists):

1. Deferred hardening (`.cursor/rules/deferred-hardening.mdc`)
2. Portal own-account move; other-bank settlement
3. Dashboard today-transaction count
4. Notify both parties on transfer

Platform learning (full roadmap):

See [TECH_LEARNING_PLAN.md](#) — rate limiting, idempotency, circuit breaker, cache, outbox, Flyway, observability, sharding lab, etc.

Redis cache-aside (local) — quick demo

1. Redis up (`docker compose`); Liberty with default `app.cache.redis-enabled=true` .
2. Clear keys: `docker exec ... redis-cli scan bankone:*` .
3. `GET /customers/{id}` or search customers twice — first miss creates `bankone:customers::...` ; second is a hit.
4. Update customer / deposit — related keys cleared.
5. Full how-it-works: [MODULES/Caching.md](#).
6. Prod / Render: keep `app.cache.redis-enabled=false` unless Redis is provisioned and eviction is reviewed.